



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
End-Spring Semester Examination 2022-23

Date of Examination: _____ Session: (FN/AN) _____ Duration: 3 hrs.

Full Marks : 50 Subject No.: NA20202

Subject : Marine Hydrodynamics

Department/Center/School: Ocean Engineering and Naval Architecture

Specific charts, graph paper, log book etc., required

Special Instructions (if any): NO

Please note that the notations and variables used in the question paper are same as that of used in the class and hence not explained here.

Answer Question no 6 and any four

1. What is circulation? Write the expression for circulation. What is Kelvin Circulation Theorem? Show that for ir-rotational flow, the circulation on a closed curve is zero. Derive the velocity potential and stream function for vortex flow. What is the complex potential for flow past a rotating cylinder? Draw an aerofoil section and define the following parameters: Leading edge, trailing edge, chord, camber.

1+1+1+1+3+1+2

2 Write the boundary value problem when a rigid body is moving in an infinite fluid domain with a velocity $V(t) = [V_1(t), V_2(t), V_3(t)]$. Show that the total force experiencing by the vessel may be written in the form:

$$F_i(t) = -\dot{U}_j m_{ij} \quad (1)$$

In equation (1), what is m_{ij} , write the expression for m_{ij} . Show that, $m_{ij} = m_{ji}$

4+3+1+1+1

3. (i) Derive the expression for incident wave potential in case of a deep water. 4

(ii) The expression for wave elevation $\eta(x, t)$ is given as $\eta(x, t) = 2 \cos(kx - 0.9t)$. Then find the value of the following parameters in case of a deep water condition. (also take $t = 0.1$ Sec, $T = 10$ Sec, density of the water as 1025 kg/m^3).

i) Wave number, ii) Wave Height, iii) Wave length, iv) Phase Velocity, v) Horizontal velocity of the water particle. vi) Vertical velocity component of the water particle

1+1+1+1+1+1

4. Show that the total energy per wave length can be written in the form of $\frac{E}{\lambda} = \frac{1}{2} \rho g A^2$. If a stone is thrown into a pond, the wave generated by that incident tends to die down at the bank. What is the reason behind?

8+2

5. Define group velocity and phase velocity. What is the physical meaning of the dispersion relation? Write the expression for the dispersion relation in the case of a finite water depth, and then derive the dispersion relation for shallow and deep water situations. Show that, unlike finite water depths, in shallow water regions, all frequency waves are travelled at the same speed. From the dispersion relation, show that for finite water depth, the group velocity and the

$$\text{phase velocity are related by the equation } C_g = \frac{1}{2} \left[C + \frac{2kh}{\sinh 2(kh)} \right]$$

1+1+2+2+4

6. (i) Show that, in the case of deep water, any object is moving along the direction of the wave propagation. **4**

(ii) Write the complete boundary value problem for linear wave structure interaction. Your answer should consist of how you define the total velocity potential, the corresponding boundary condition, and the equation of motion. **4**

(iii) Define A_{ij} and B_{ij} . These two components are part of which force? **2**

----- All the best -----